Multimodal data integration study

To do | In progress | Done | Not concerned

PROJECT SETUP

Human Resources

- Project manager
- Clinician/Biologist
- Biostatistician/Bioinformatician/Computational Biologist
- Data steward
- Ethicist/Legal expert

Computer Resources

- HPC/GPU access
- Short-term storage
- Long-term storage
- Energy footprint assessed

Good practices

- FAIR principles
- CARE principles
- Open code/data policy defined
- Licences

INTEGRATION PROTOCOL

RESEARCH QUESTION

Formulation

- Precise biological/clinical question
- Outcomes and measures specified
- Success criteria defined

Scope

- Population/system defined
- Rationale for multimodality
- Exploratory or inferential objectives
- Interpretability requirement
- Sample size/Power/Biological replicates

DATA PREPARATION

Governance

- Regulatory compliance and re-identification risk
- Access, authorisations and consent
- Metadata documented
- Data Management Plan

Per-modality preparation

- Modalities to collect and/or reuse defined
- QC and preprocessing documented
- Confounders charaterised (batch, bias ...)
- Exploratory analysis

Cross-modal connections

- Connection type (vertical/horizontal ...)
- Missing modality pattern
- Redundancy/complementarity
- Data harmonisation

ALGORITHM SELECTION

Task formulation

- Task type defined
- Learning strategy chosen
- Algorithm assumptions checked against data

Multimodal integration

- Integration timing (early/intermediate/late)
- Intermediate fusion approach (if applicable)
- Prior resources incorporated

Practical constraints

- Interpretability approach defined
- Computational feasibility
- Software availability and maintenance

EVALUATION

Validation

- Internal validation strategy
- External validation and representativeness
- Data leakage risks controled

Metrics

- Metrics suited to task
- Unimodal baseline established
- Clinical utility considered

Multimodal evaluation

- Modality ablation and dropout
- Cross-modal validation
- Modality importance

Reproducibility

- Version control (data, script ...)
- Workflow manager
- Documentation
- Benchmark/data challenge
- Pre-registration